

Chapter 04: Derivations

Derivation 1: Solve the following differential equations for the upward (F^u) and downward (F^d) fluxes for a canopy with horizontal leaves,

$$\frac{\partial}{\partial L} F^d(L) = \rho_L F^u(L) + (\tau_L - 1) F^d(L) , \quad (1a)$$

$$-\frac{\partial}{\partial L} F^u(L) = \rho_L F^d(L) + (\tau_L - 1) F^u(L) . \quad (1b)$$

Derivation 2: Show the explicit form for F^u in the limiting case of a very dense canopy.

Derivation 3: Derive expressions for the coefficients K_1^d , K_2^d , K_3^d , K_1^u , K_2^u and K_3^u in the case of a canopy with uniform leaf normal orientation ($G = 0.5$) and bi-Lambertian leaf scattering only [Γ given by Eq. (2.22)]. For definitions of these coefficients see § IV.8.

Derivation 4: For the uniform leaf normal and rotationally invariant scattering kernel problem in Derivation 3, for which you derived the coefficients, solve the differential equations for the upward (F^u) and downward (F^d) fluxes. Again, derive the expression for the upward (F^u) flux in the limiting case of a very dense canopy.

Derivation 5: Derive the two-stream transport equations given in Sellers (Canopy reflectance, photosynthesis and transpiration. Int. J. Remote Sens., 6:1335-1372, 1985) starting with the radiative transfer equation.

Chapter 04: Derivation 1 Answer:

For a horizontally homogeneous leaf canopy consisting of horizontal leaves, we have the simplified differential radiative transfer equations for the upward (F^u) and downward (F^d) fluxes

$$\begin{cases} \frac{\partial}{\partial L} F^d(L) = \rho_L F^u(L) + (\tau_L - 1) F^d(L) \\ -\frac{\partial}{\partial L} F^u(L) = \rho_L F^d(L) + (\tau_L - 1) F^u(L) \end{cases} \quad (1)$$

with the boundary conditions

$$\begin{cases} F^d(L = 0) = F_0^d \\ F^u(L = L_H) = r_s F^d(L = L_H) \end{cases} \quad (2)$$

Our goal is to get the analytical solution of Eq. (1) using four parameters of ρ_L , τ_L , r_s , and F_0^d .

We solve first equation of Eqs.(1) for $F_d(L)$ and substitute into the second equation to obtain the differential equation

$$\frac{d^2 F^d(L)}{dL^2} - \alpha^2 F^d(L) = 0 \quad (3)$$

The differential equation

$$\frac{d^2 F^u(L)}{dL^2} - \alpha^2 F^u(L) = 0 \quad (4)$$

can be adjoined by analogy. The constant α in Eqs. (3) and (4) is specified by the relation

$$\alpha = [(1 - \tau_L)^2 - \rho_L^2]^{1/2} \quad (5)$$

General solution of Eqns. (3) and (4) are

$$F^d(L) = C_1 e^{\alpha L} + C_2 e^{-\alpha L} \quad (6)$$

and

$$F^u(L) = C_3 e^{\alpha L} + C_4 e^{-\alpha L} \quad (7)$$

respectively, where C_1 , C_2 , C_3 and C_4 are constants, which are to be specified by using boundary conditions

$$\left\{ \begin{array}{l} F^d(L=0) = F_0^d \\ F^u(L=0) = F_0^u \\ \frac{\partial F^d(L)}{\partial L}(L=0) = \rho_L F_0^u + (\tau_L - 1)F_0^d \\ \frac{\partial F^u(L)}{\partial L}(L=0) = -\rho_L F_0^d - (\tau_L - 1)F_0^u \\ F^u(L=L_H) = r_s F^d(L=L_H) \end{array} \right. \quad (8)$$

Plugging Eqs. (6) and (7) into (8) we obtain

$$\left\{ \begin{array}{l} C_1 + C_2 = F_0^d \\ C_3 + C_4 = F_0^u \\ \alpha(C_1 - C_2) = \rho_L F_0^u + (\tau_L - 1)F_0^d \\ \alpha(C_3 - C_4) = -\rho_L F_0^d - (\tau_L - 1)F_0^u \\ C_3 e^{\alpha L_H} + C_4 e^{-\alpha L_H} = r_s (C_1 e^{\alpha L_H} + C_2 e^{-\alpha L_H}) \end{array} \right. \quad (9)$$

The solutions of these equations are

$$\left\{ \begin{array}{l} C_1 = \frac{\rho_L F_0^u - (1 - \tau_L - \alpha)F_0^d}{2\alpha} \\ C_2 = \frac{(1 - \tau_L + \alpha)F_0^d - \rho_L F_0^u}{2\alpha} \\ C_3 = \frac{(1 - \tau_L + \alpha)F_0^u - \rho_L F_0^d}{2\alpha} \\ C_4 = \frac{\rho_L F_0^d - (1 - \tau_L - \alpha)F_0^u}{2\alpha} \\ F_0^u = \frac{b_1 e^{-2\alpha L_H} - b_2}{a_1 - a_2 e^{-2\alpha L_H}} F_0^d \end{array} \right. \quad (10)$$

where $a_1 = r_s - \frac{1}{A'}$, $a_2 = r_s - A'$, $b_1 = 1 - \frac{r_s}{A'}$, $b_2 = 1 - r_s A'$, and $A' = \frac{1 - \tau_L - \alpha}{\rho_L} = \frac{\rho_L}{1 - \tau_L + \alpha}$.

Then we get our final solutions of Eqns. (1) as following

$$\left\{ \begin{array}{l} F^d(L) = F_0^d \frac{a_1 e^{-\alpha L} - a_2 e^{-\alpha(2L_H - L)}}{a_1 - a_2 e^{-2\alpha L_H}} \\ F^u(L) = F_0^u \frac{b_1 e^{-\alpha(2L_H - L)} - b_2 e^{-\alpha L}}{a_1 - a_2 e^{-2\alpha L_H}} \end{array} \right. \quad (11)$$

Chapter 04: Derivation 2 Answer:

For upward (F^u) flux of a canopy with horizontal leaves, we have

$$F^u(L) = F_0^d \frac{b_1 e^{-\alpha(2L_H - L)} - b_2 e^{-\alpha L}}{a_1 - a_2 e^{-2\alpha L_H}} \quad (1)$$

In the case of a very dense canopy, $L_H \rightarrow \infty$. So we have

$$F^u(L) = F_0^d \frac{0 - b_2 e^{-\alpha L}}{a_1 - 0} \quad (2)$$

Note that $a_1 = r_s - \frac{1}{A'}$ and $b_2 = 1 - r_s A'$. So we have

$$\begin{aligned} F^u(L) &= F_0^d \frac{-b_2 e^{-\alpha L}}{a_1} \\ &= F_0^d \frac{(r_s A' - 1) e^{-\alpha L}}{r_s - \frac{1}{A'}} \\ &= F_0^d A' e^{-\alpha L} \end{aligned} \quad (3)$$

Derivation 3:

Derive expressions for the coefficients K_1^d , K_2^d , K_3^d , K_1^u , K_2^u , K_3^u in the case of a canopy with uniform leaf normal orientation ($G = 0.5$) and bi-Lambertian leaf scattering only.

Starting from general radiative transfer equation:

$$-\mu \frac{\partial I(L, \Omega)}{\partial L} + G(\Omega) I(L, \Omega) = \frac{1}{\pi} \int_{4\pi} \Gamma(\Omega' \rightarrow \Omega) I(L, \Omega') d\Omega'. \quad (1)$$

The *normalized scattering phase function* $(1/4\pi)P$ for diffuse scattering only is

$$P(\Omega' \rightarrow \Omega) = \frac{4\Gamma(\Omega' \rightarrow \Omega)}{\omega_L G(\Omega')}.$$

Rewriting equation (1) as

$$-\mu \frac{\partial I(L, \mu, \phi)}{\partial L} + G(\mu) I(L, \mu, \phi) = \frac{\omega_L}{4\pi} \int_{-1}^1 \int_0^{2\pi} P(\mu, \phi; \mu', \phi') G(\mu') I(L, \mu', \phi') d\mu' d\phi'. \quad (2)$$

In the case of uniformly distributed leaf normals, $G(\Omega)$ is 0.5, and the area scattering phase function and hence $P(\mu, \phi; \mu', \phi')$ is rotationally invariant:

$$\Gamma(\Omega' \rightarrow \Omega) = \frac{\omega_L}{3\pi} (\sin \beta - \beta \cos \beta) + \frac{\tau_L}{2} \cos \beta, \quad (3)$$

where $\beta = \arccos(\Omega' \bullet \Omega) = \arccos\{\sin \theta \sin \theta' \cos(\phi - \phi') + \cos \theta \cos \theta'\}$.

For phase function as a function only of the scattering angle, Chandrasekhar (1960) has shown that the azimuthal integral satisfies

$$p(\mu, \mu') = \frac{1}{2\pi} \int_0^{2\pi} P(\mu, \phi; \mu', \phi') d\phi,$$

With two point quadrature approximation

$$\int_{-1}^1 p(\mu, \mu') I(L, \mu') d\mu' = p(\mu, \mu_1) I(L, \mu_1) + p(\mu, -\mu_1) I(L, -\mu_1),$$

and the additional definition of azimuthal averaged intensity $I(L, \mu)$

$$I(L, \mu) = \int_0^{2\pi} I(L, \mu, \phi) d\phi,$$

we can get a simpler expression for the hemispherical integration of the rightside term of the RT equation (2),

$$\begin{aligned}
& \frac{\omega_L}{8\pi} \int_{-1}^0 d\mu \int_0^{2\pi} d\phi \int_{-1}^1 \int_0^{2\pi} P(\mu, \phi; \mu', \phi') I(L, \mu', \phi') d\mu' d\phi' \\
&= \frac{\omega_L}{8\pi} \int_{-1}^0 d\mu \int_{-1}^1 d\mu' \int_0^{2\pi} I(L, \mu', \phi') d\phi' \int_0^{2\pi} P(\mu, \phi; \mu', \phi') d\phi \\
&= \frac{\omega_L}{4} \int_{-1}^0 d\mu \int_{-1}^1 p(\mu, \mu') d\mu' \int_0^{2\pi} I(L, \mu', \phi') d\phi' \\
&= \frac{\omega_L}{4} \int_{-1}^0 d\mu \int_{-1}^1 p(\mu, \mu') I(L, \mu') d\mu' \\
&= \frac{\omega_L}{4} \int_{-1}^0 \{p(\mu, \mu_1) I(L, \mu_1) + p(\mu, -\mu_1) I(L, -\mu_1)\} d\mu \\
&= \frac{\omega_L}{4} \{I(L, \mu_1) \int_{-1}^0 p(\mu, \mu_1) d\mu + I(L, -\mu_1) \int_{-1}^0 p(\mu, -\mu_1) d\mu\}.
\end{aligned}$$

To further simplify the above term, define the quantity β_i

$$\begin{aligned}
\beta_i &= \frac{1}{2} \int_{-1}^0 p(\mu_i, \mu') d\mu' \\
&= 1 - \frac{1}{2} \int_0^1 p(\mu_i, \mu') d\mu',
\end{aligned}$$

and make the following quadrature-like approximation for downwelling flux $F^d(L)$ and upwelling flux $F^u(L)$

$$\begin{aligned}
F^d(L) &= \int_{-1}^0 |\mu| I(L, \mu) d\mu \\
&\simeq \mu_1 I(L, -\mu_1), \\
F^u(L) &= \int_0^1 |\mu| I(L, \mu) d\mu \\
&\simeq \mu_1 I(L, \mu_1),
\end{aligned}$$

indicating the following approximation

$$\begin{aligned}
I(L, \mu_1) &\simeq F^u(L)/\mu_1, \\
I(L, -\mu_1) &\simeq F^d(L)/\mu_1.
\end{aligned}$$

So we get simplified expression,

$$\frac{\omega_L}{8\pi} \{I(L, \mu_1) \int_{-1}^0 p(\mu, \mu_1) d\mu + I(L, -\mu_1) \int_{-1}^0 p(\mu, -\mu_1) d\mu\} = \frac{\omega_L}{2\mu_1} \{(1-\beta_1)F^d(L) + \beta_1 F^u(L)\}.$$

Now, applying the operator $\int_{2\pi-} d\Omega$ to equation (2),

$$\frac{\partial}{\partial L} \int_{-1}^0 \{(-\mu) I(L, \mu)\} d\phi + \frac{1}{2} \int_{-1}^0 I(L, \mu) d\mu = \frac{\omega_L}{8\pi} \int_{-1}^0 d\mu \int_0^{2\pi} d\phi \int_{-1}^1 \int_0^{2\pi} P(\mu, \phi; \mu', \phi') I(L, \mu', \phi') d\mu' d\phi',$$

$$\frac{\partial F^d(L)}{\partial L} + \frac{1}{2\mu_1} F^d(L) = \frac{\omega_L}{2\mu_1} \{(1 - \beta_1) F^d(L) + \beta_1 F^u(L)\}. \quad (4)$$

Similarly, for the upwelling flux, applying the operator $\int_{2\pi+} d\Omega$ to equation (2), we can get

$$-\frac{\partial}{\partial L} \int_0^1 \{(\mu) I(L, \mu)\} d\mu + \frac{1}{2} \int_0^1 I(L, \mu) d\mu = \frac{\omega_L}{8\pi} \int_0^1 d\mu \int_0^{2\pi} d\phi \int_{-1}^1 \int_0^{2\pi} P(\mu, \phi; \mu', \phi') I(L, \mu', \phi') d\mu' d\phi',$$

$$-\frac{\partial F^d(L)}{\partial L} + \frac{1}{2\mu_1} F^u(L) = \frac{\omega_L}{4} \{I(L, \mu_1) \int_0^1 p(\mu, \mu_1) d\mu + I(L, -\mu_1) \int_0^1 p(\mu, -\mu_1) d\mu\},$$

$$-\frac{\partial F^d(L)}{\partial L} + \frac{1}{2\mu_1} F^u(L) = \frac{\omega_L}{2\mu_1} \{\beta_1 F^d(L) + (1 - \beta_1) F^u\}. \quad (5)$$

Rearranging equation (4) and (5) as

$$\frac{\partial F^d(L)}{\partial L} = -\frac{1}{2\mu_1} [1 - \omega_L(1 - \beta_1)] F^d(L) + \frac{\omega_L \beta_1}{2\mu_1} F^u(L), \quad (6)$$

$$\frac{\partial F^u(L)}{\partial L} = -\frac{\omega_L \beta_1}{2\mu_1} F^d(L) + \frac{1}{2\mu_1} [1 - \omega_L(1 - \beta_1)] F^u(L). \quad (7)$$

With Gaussian choice, $\mu_1 = \frac{1}{\sqrt{3}}$.

The only unknown parameter is β_1 , so the key then is to derive β_1 with *normalized scattering phase function* inferred from equation (3) for uniformly distributed leaf normal,

$$\begin{aligned} p(\mu, \mu') &= \frac{1}{2\pi} \int_0^{2\pi} P(\mu, \phi; \mu', \phi') d\phi \\ &= \frac{4}{2\pi\omega_L 0.5} \int_0^{2\pi} \left\{ \frac{\omega_L}{3\pi} (\sin \beta - \beta \cos \beta) + \frac{\tau_L}{2} \cos \beta \right\} d\phi \\ &= \frac{4}{\pi\omega_L} \int_0^{2\pi} \left\{ \frac{\tau_L}{2} - \frac{\omega_L}{3\pi} \right\} \cos \beta d\phi + \frac{4}{3\pi^2} \int_0^{2\pi} \sin \beta d\phi \\ &= \frac{4}{\pi\omega_L} \left\{ \frac{\tau_L}{2} - \frac{\omega_L}{3\pi} \right\} \int_0^{2\pi} \{\sin \theta \sin \theta' \cos(\phi - \phi') + \cos \theta \cos \theta'\} d\phi + \frac{4}{3\pi^2} \int_0^{2\pi} \sin \beta d\phi \\ &= \frac{4}{\pi\omega_L} \left\{ \frac{\tau_L}{2} - \frac{\omega_L}{3\pi} \right\} \cos \theta \cos \theta' + \frac{4}{3\pi^2} \int_0^{2\pi} \sin \beta d\phi, \end{aligned}$$

$$\begin{aligned} \beta_1 &= \frac{1}{2} \int_{-1}^0 p(\mu_1, \mu') d\mu' \\ &= \frac{1}{2} \frac{4}{\pi\omega_L} \left\{ \frac{\tau_L}{2} - \frac{\omega_L}{3\pi} \right\} \int_{-1}^0 \cos \theta_1 \cos \theta' d\mu' + \frac{1}{2} \int_{-1}^0 \frac{4}{3\pi^2} d\mu' \int_0^{2\pi} \sin \beta d\phi \\ &= \frac{2\mu_1}{\pi\omega_L} \left\{ \frac{\tau_L}{2} - \frac{\omega_L}{3\pi} \right\} \frac{1}{2} + \frac{2}{3\pi^2} \int_{-1}^0 d\mu' \int_0^{2\pi} \sin \beta d\phi. \end{aligned}$$

Since the last term is relatively small, the following approximation could be made

$$\beta_1 \simeq \frac{\mu_1}{\pi\omega_L} \left\{ \frac{\tau_L}{2} - \frac{\omega_L}{3\pi} \right\}.$$

Derivation 4:

For the uniform leaf normal and rotationally invariant scattering kernel problem in Derivation 3, for which you derived the coefficients, solve the differential equations for the upward (F^u) and downward (F^d) fluxes. Again, derive the expression for the upward (F^u) flux in the limiting case of a very dense canopy.

Solving the linear differential-integral equations (6) and (7) with constant coefficients:

$$\frac{\partial F^d(L)}{\partial L} = C_1 F^d(L) + C_2 F^u(L), \quad (8)$$

$$\frac{\partial F^u(L)}{\partial L} = C_3 F^d(L) + C_4 F^u(L), \quad (9)$$

where

$$\begin{aligned} C_1 &= -\frac{1}{2\mu_1}[1 - \omega_L(1 - \beta_1)] = -C_4, \\ C_2 &= \frac{\omega_L\beta_1}{2\mu_1}, \\ C_3 &= -\frac{\omega_L\beta_1}{2\mu_1} = -C_2, \\ C_4 &= \frac{1}{2\mu_1}[1 - \omega_L(1 - \beta_1)], \end{aligned} \quad (10)$$

$$\text{and } \beta_1 = \frac{\mu_1}{\pi\omega_L}\left\{\frac{\tau_L}{2} - \frac{\omega_L}{3\pi}\right\}, \mu_1 = \frac{1}{\sqrt{3}}.$$

The boundary conditions are

$$F^d(L=0) = F^0; F^u(L=L_H) = \alpha F^d(L=L_H). \quad (11)$$

where α is surface *albedo* and F^0 is the incident solar flux on the upper boundary of canopy.

From characterization matrix equation

$$\begin{vmatrix} C_1 - \lambda & C_2 \\ C_3 & C_4 - \lambda \end{vmatrix} = 0,$$

we get

$$\begin{aligned} \lambda_{1,2} &= \pm \sqrt{C_2 C_3 - C_1 C_4} \\ &= \pm \frac{1}{2\mu_1} \sqrt{[1 - \omega_L(1 - \beta_1)]^2 - (\omega\beta_1)^2} \\ &= \pm \frac{1}{2\mu_1} \sqrt{1 - \omega} \sqrt{1 - \omega + 2\omega\beta_1}. \end{aligned}$$

So, $F_d(L)$ and $F^u(L)$ have following gernerl forms of solution,

$$\begin{aligned} F^d(L) &= A_1 e^{-\lambda L} + B_1 e^{\lambda L}, \\ F^u(L) &= A_2 e^{-\lambda L} + B_2 e^{\lambda L}, \end{aligned} \quad (12)$$

$$\text{where } \lambda = \frac{1}{2\mu_1} \sqrt{1 - \omega} \sqrt{1 - \omega + 2\omega\beta_1}.$$

Plugging equations (12) to equations (8) and (9), the A_1, B_1, A_2 and B_2 are constrained by

$$\begin{aligned} (\lambda + C_1)A_1 &= -C_2 A_2, \\ (\lambda - C_1)A_1 &= C_2 B_2. \end{aligned} \quad (13)$$

Combined with two boundary conditions :

$$\begin{aligned} F^d(L=0) &= A_1 + B_1 = F^0, \\ F^u(L=L_H) &= A_2 e^{-\lambda L_H} + B_2 e^{\lambda L_H} = \alpha F_d(L=L_H) = \alpha \{A_1 e^{-\lambda L_H} + B_1 e^{\lambda L_H}\}. \end{aligned} \quad (14)$$

Solving A_1, B_1, A_2 and B_2 from the above four equations (13) and (14), we get

$$\begin{aligned} A_1 &= \frac{\lambda - C_1 - C_2 \alpha}{(\lambda - C_1 - C_2 \alpha) + (\lambda + C_1 + C_2 \alpha) \exp(-2\lambda L_H)} F^0, \\ A_2 &= \frac{(\lambda + C_1) \alpha + C_2}{(\lambda - C_1 - C_2 \alpha) + (\lambda + C_1 + C_2 \alpha) \exp(-2\lambda L_H)} F^0, \\ B_1 &= \frac{\lambda + C_1 + C_2 \alpha}{(\lambda + C_1 + C_2 \alpha) + (\lambda - C_1 - C_2 \alpha) \exp(2\lambda L_H)} F^0, \\ B_2 &= \frac{(\lambda - C_1) \alpha - C_2}{(\lambda + C_1 + C_2 \alpha) + (\lambda - C_1 - C_2 \alpha) \exp(2\lambda L_H)} F^0. \end{aligned} \quad (15)$$

λ , C_1 and C_2 all have $\frac{1}{2\mu_1}$, and A_1, B_1, A_2, B_2 are all linear combinations of these three parameters, so $\frac{1}{2\mu_1}$ can be dropped simultaneously when calculating downwelling $F^d(L)$ and upwelling flux $F^u(L)$.

The final solutions of $F^d(L)$ and $F^u(L)$ are expression (12) with the above resolved coefficients. In the case of black soil ($\alpha = 0$), the four coefficients in the solutions can be simplified as

$$\begin{aligned} A_1 &= \frac{\lambda - C_1}{(\lambda - C_1) + (\lambda + C_1) \exp(-2\lambda L_H)} F^0, \\ A_2 &= \frac{C_2}{(\lambda - C_1) + (\lambda + C_1) \exp(-2\lambda L_H)} F^0, \\ B_1 &= \frac{\lambda + C_1}{(\lambda + C_1) + (\lambda - C_1) \exp(2\lambda L_H)} F^0, \\ B_2 &= \frac{-C_2}{(\lambda + C_1) + (\lambda - C_1) \exp(2\lambda L_H)} F^0. \end{aligned} \quad (16)$$

In the limiting case of a very dense canopy ($L_H \rightarrow \infty$),

$$\begin{aligned} A_1 &= F^0, \\ A_2 &= \frac{(\lambda + C_1) \alpha + C_2}{(\lambda - C_1 - C_2 \alpha)} F^0, \\ B_1 &= 0, \\ B_2 &= 0. \end{aligned} \quad (17)$$

From equation (11), the upwelling flux is

$$\begin{aligned} F^u(L) &= A_2 e^{-\lambda L} + B_2 e^{\lambda L}, \\ &= A_2 e^{-\lambda L}, \\ &= \frac{(\lambda + C_1) \alpha + C_2}{(\lambda - C_1 - C_2 \alpha)} F^0 e^{-\lambda L}, \\ &= \frac{(\sqrt{1 - \omega_L} \sqrt{1 - \omega_L + 2\omega_L \beta_1}) \alpha + \omega_L \beta_1}{\sqrt{1 - \omega_L} \sqrt{1 - \omega_L + 2\omega_L \beta_1} + [1 - \omega_L (1 - \beta_1)] - \omega_L \beta_1 \alpha} F^0 e^{-\sqrt{1 - \omega_L} \sqrt{1 - \omega_L + 2\omega_L \beta_1} L / (2\mu_1)}. \end{aligned}$$

I. One-angle form of radiative transfer equation

The radiative transfer equation for vegetation canopies is

$$-\mu \frac{\partial}{\partial z} I(z, \Omega) + u_L(z) G(z, \Omega) I(z, \Omega) = \frac{u_L(z)}{\pi} \int_{4\pi} \Gamma(z, \Omega' \rightarrow \Omega) I(z, \Omega') d\Omega' . \quad (1)$$

Dividing the above equation through by the *leaf area density distribution* $u_L(z)$, we can change the vertical coordinate from *depth* z to *cumulative leaf area index* L , namely,

$$-\mu \frac{\partial}{\partial L} I(L, \Omega) + G(L, \Omega) I(L, \Omega) = \frac{1}{\pi} \int_{4\pi} \Gamma(L, \Omega' \rightarrow \Omega) I(L, \Omega') d\Omega' . \quad (2)$$

Assume the angular distribution of leaves, $G(L, \Omega)$ and $\Gamma(L, \Omega' \rightarrow \Omega)$, are independent of the *azimuth angle* (ϕ, ϕ') . The above equation can be reduced to a one-angle problem by simply averaging over the *azimuth angle* ϕ :

$$\frac{1}{2\pi} \int_0^{2\pi} \left[-\mu \frac{\partial}{\partial L} I(L, \Omega) \right] d\phi = -\mu \frac{\partial}{\partial L} \frac{1}{2\pi} \int_0^{2\pi} I(L, \phi, \mu) d\phi = -\mu \frac{\partial}{\partial L} I(L, \mu) ,$$

$$\begin{aligned} \frac{1}{2\pi} \int_0^{2\pi} [G(L, \Omega) I(L, \Omega)] d\phi &= \frac{1}{2\pi} \int_0^{2\pi} G(L, \phi, \mu) I(L, \phi, \mu) d\phi \\ &= G(L, \mu) \frac{1}{2\pi} \int_0^{2\pi} I(L, \phi, \mu) d\phi = G(L, \mu) I(L, \mu) , \end{aligned}$$

$$\begin{aligned} \frac{1}{2\pi} \int_0^{2\pi} \left[\int_{4\pi} \Gamma(L, \Omega' \rightarrow \Omega) I(L, \Omega') d\Omega' \right] d\phi &= \frac{1}{2\pi} \int_0^{2\pi} d\phi \int_{-1}^1 d\mu' \int_0^{2\pi} \Gamma(L, \phi' \rightarrow \phi, \mu' \rightarrow \mu) I(L, \phi', \mu') d\phi' \\ &= \frac{1}{2\pi} \int_0^{2\pi} d\phi \int_{-1}^1 d\mu' \Gamma(L, \mu' \rightarrow \mu) \int_0^{2\pi} I(L, \phi', \mu') d\phi' = \int_0^{2\pi} d\phi \int_{-1}^1 d\mu' \Gamma(L, \mu' \rightarrow \mu) I(L, \mu') \\ &= 2\pi \int_{-1}^1 d\mu' \Gamma(L, \mu' \rightarrow \mu) I(L, \mu') , \end{aligned}$$

$$\text{where } I(L, \mu) = \frac{1}{2\pi} \int_0^{2\pi} I(L, \phi, \mu) d\phi .$$

So one-angle form of equation (2) is given by

$$-\mu \frac{\partial}{\partial L} I(L, \mu) + G(L, \mu) I(L, \mu) = 2 \int_{-1}^1 \Gamma(L, \mu' \rightarrow \mu) I(L, \mu') d\mu' . \quad (3)$$

Let assume the angular distribution of leaves, $G(L, \Omega)$ and $\Gamma(L, \Omega' \rightarrow \Omega)$, are independent of the *cumulative leaf area index* L . Equation (3) can be simply reduced to

$$-\mu \frac{\partial}{\partial L} I(L, \mu) + G(\mu) I(L, \mu) = 2 \int_{-1}^1 \Gamma(\mu' \rightarrow \mu) I(L, \mu') d\mu', \quad (4)$$

with the boundary conditions,

$$I(L = 0, \mu) = I_0 \delta(\mu - \mu_0) + I_d(\mu), \mu < 0, \quad (5)$$

$$I(L = L_H, \mu) = 2 \int_0^1 d\mu' |\mu'| \rho_s(\mu' \rightarrow \mu) I(L = L_H, \mu'), \mu > 0, \mu' < 0, \quad (6)$$

where

$$I_0 = \frac{f_{\text{dir}}}{|\mu_0|} F_L(L = 0) = f_{\text{dir}} S_0, \quad S_0 = F_L(L = 0)/|\mu_0|, \text{ and,}$$

$$I_d = (1 - f_{\text{dir}}) d_0(L = 0, \mu) F_L(L = 0) = (1 - f_{\text{dir}}) S_d, \quad S_d = d_0(L = 0, \mu) F_L(L = 0).$$

II. Two-stream approximation of radiative transfer model

1. The $I(L, \mu)$ can be separated into the uncollided and collided components

$$I(L, \mu) = I^o(L, \mu) + I^c(L, \mu). \quad (7)$$

The transport equation (4) can be split into equation for the uncollided problem

$$-\mu \frac{\partial}{\partial L} I^o(L, \mu) + G(\mu) I^o(L, \mu) = 0, \quad (8)$$

$$I^o(L = 0, \mu) = f_{\text{dir}} S_0 \delta(\mu - \mu_0) + (1 - f_{\text{dir}}) S_d, \mu < 0, \quad (9)$$

$$I^o(L = L_H, \mu) = 2 \int_0^1 d\mu' |\mu'| \rho_s(\mu' \rightarrow \mu) I^o(L = L_H, \mu'), \mu > 0, \mu' < 0, \quad (10)$$

and collided problem

$$-\mu \frac{\partial}{\partial L} I^c(L, \mu) + G(\mu) I^c(L, \mu) = 2 \int_{-1}^1 \Gamma(\mu' \rightarrow \mu) [I^o(L, \mu') + I^c(L, \mu')] d\mu', \quad (11)$$

$$I^c(L = 0, \mu) = 0, \mu < 0, \quad (12)$$

$$I^c(L = L_H, \mu) = 2 \int_0^1 d\mu' |\mu'| \rho_s(\mu' \rightarrow \mu) I^c(L = L_H, \mu'), \mu > 0, \mu' < 0. \quad (13)$$

2. The uncollided problem

Let $K = G(\mu)/\mu$, equation (8) can be written as

$$\frac{\partial}{\partial L} I^o(L, \mu) = K I^o(L, \mu), \quad (14)$$

with the boundary conditions (9) and (10). We get the solution

$$I^o(L, \mu) = [f_{\text{dir}} S_0 \delta(\mu - \mu_0) + (1 - f_{\text{dir}}) S_d] \exp(KL), \mu < 0, \quad (15)$$

$$\begin{aligned} I^o(L, \mu) &= 2 \int_0^1 d\mu' |\mu' | \rho_s(\mu' \rightarrow \mu) I^o(L = L_H, \mu') \exp(K(L - L_H)) \\ &= 2 \int_0^1 d\mu' |\mu' | \rho_s(\mu' \rightarrow \mu) [f_{\text{dir}} S_0 \delta(\mu - \mu_0) + (1 - f_{\text{dir}}) S_d] \exp(G(\mu') L_H / \mu') \exp(K(L - L_H)) \\ &= 2 |\mu_0 | \rho_s(\mu_0 \rightarrow \mu) f_{\text{dir}} S_0 \exp(G(\mu_0) L_H / \mu_0) \exp(K(L - L_H)) + 2(1 - f_{\text{dir}}) S_d \\ &\quad \left[\int_0^1 |\mu' | \rho_s(\mu' \rightarrow \mu) \Gamma(\mu' \rightarrow \mu) \exp(G(\mu') L_H / \mu') d\mu' \right] \exp(K(L - L_H)), \mu > 0, \mu' < 0. \end{aligned} \quad (16)$$

For the dense canopy, we simplify equation (16) as

$$I^o(L, \mu) = 0, \mu > 0. \quad (17)$$

3. The collided problem

The right hand in the collided problem (11) can be rewritten as

$$2 \int_{-1}^1 \Gamma(\mu' \rightarrow \mu) [I^o(L, \mu') + I^c(L, \mu')] d\mu' = 2 \int_{-1}^1 \Gamma(\mu' \rightarrow \mu) I^o(L, \mu') d\mu' + 2 \int_{-1}^1 \Gamma(\mu' \rightarrow \mu) I^c(L, \mu') d\mu'.$$

Let consider the dense vegetation for the first component in the right hand

$$\begin{aligned} \int_{-1}^1 \Gamma(\mu' \rightarrow \mu) I^o(L, \mu') d\mu' &= \int_{-1}^1 \Gamma(\mu' \rightarrow \mu) [f_{\text{dir}} S_0 \delta(\mu' - \mu_0) + (1 - f_{\text{dir}}) S_d] \exp(G(\mu') L / \mu') d\mu' \\ &= f_{\text{dir}} S_0 \Gamma(\mu_0 \rightarrow \mu) \exp(G(\mu_0) L / \mu_0) + (1 - f_{\text{dir}}) S_d \int_{-1}^1 \Gamma(\mu' \rightarrow \mu) \exp(G(\mu') L / \mu') d\mu'. \end{aligned}$$

The collided equation (11) becomes

$$\begin{aligned} -\mu \frac{\partial}{\partial L} I^c(L, \mu) + G(\mu) I^c(L, \mu) &= 2 f_{\text{dir}} S_0 \Gamma(\mu_0 \rightarrow \mu) \exp(G(\mu_0) L / \mu_0) \\ &\quad + 2(1 - f_{\text{dir}}) S_d \int_{-1}^1 \Gamma(\mu' \rightarrow \mu) \exp(G(\mu') L / \mu') d\mu' + 2 \int_{-1}^1 \Gamma(\mu' \rightarrow \mu) I^c(L, \mu') d\mu', \end{aligned} \quad (18)$$

and with the boundary condition (12) and (13).

4. Approximate solution for diffuse contribution

Let assume the canopy is vertically homogeneous. Multiply equation (18) by $|\mu|/G(\mu)$ and integrate μ over $[-1, 0], [0, 1]$, respectively,

$$\int_0^1 \left[G(\mu) \frac{|\mu|}{G(\mu)} I^c(L, \mu) \right] d\mu = \int_0^1 \mu I^c(L, \mu) d\mu = I^c(L) \uparrow,$$

$$\int_{-1}^0 \left[G(\mu) \frac{|\mu|}{G(\mu)} I^c(L, \mu) \right] d\mu = \int_{-1}^0 |\mu| I^c(L, \mu) d\mu = I^c(L) \downarrow,$$

$$\int_0^1 \left[-\mu \frac{|\mu|}{G(\mu)} \frac{\partial}{\partial L} I^c(L, \mu) \right] d\mu = -\bar{\mu} \frac{\partial}{\partial L} I^c(L) \uparrow,$$

$$\int_{-1}^0 \left[-\mu \frac{|\mu|}{G(\mu)} \frac{\partial}{\partial L} I^c(L, \mu) \right] d\mu = \bar{\mu} \frac{\partial}{\partial L} I^c(L) \downarrow,$$

$$\text{where } \bar{\mu} = \frac{\int_0^1 \left[\mu \frac{|\mu|}{G(\mu)} I^c(L, \mu) \right] d\mu}{\int_0^1 \mu I^c(L, \mu) d\mu},$$

$$\begin{aligned} & \int_0^1 \left[\frac{|\mu|}{G(\mu)} 2 \int_{-1}^1 \Gamma(\mu' \rightarrow \mu) I^o(L, \mu') d\mu' \right] d\mu \\ &= \int_0^1 \left[\frac{|\mu|}{G(\mu)} 2 \int_{-1}^1 \Gamma(\mu' \rightarrow \mu) [f_{\text{dir}} S_0 \delta(\mu' - \mu_0) + (1 - f_{\text{dir}}) S_d] \exp(K' L) d\mu' \right] d\mu \\ &= 2 f_{\text{dir}} S_0 \exp(G(\mu_0) L / \mu_0) \int_0^1 \Gamma(\mu_0 \rightarrow \mu) \frac{\mu}{G(\mu)} d\mu \\ &+ 2(1 - f_{\text{dir}}) S_d \int_0^1 \exp(G(\mu') L / \mu') d\mu' \int_0^1 \Gamma(\mu' \rightarrow \mu) \frac{\mu}{G(\mu)} d\mu \\ &= f_{\text{dir}} S_0 \exp(G(\mu_0) L / \mu_0) \gamma_0^+(\mu_0) + (1 - f_{\text{dir}}) S_d \gamma_d^+(L), \end{aligned}$$

$$\text{Where } \gamma_0^+(\mu_0) = 2 \int_0^1 \Gamma(\mu_0 \rightarrow \mu) \frac{\mu}{G(\mu)} d\mu,$$

$$\gamma_d^+(L) = 2 \int_0^1 \exp(G(\mu') L / \mu') d\mu' \int_0^1 \Gamma(\mu' \rightarrow \mu) \frac{\mu}{G(\mu)} d\mu$$

$$\begin{aligned}
& \int_{-1}^0 \left[\frac{|\mu|}{G(\mu)} 2 \int_{-1}^1 \Gamma(\mu' \rightarrow \mu) I^o(L, \mu') d\mu' \right] d\mu \\
&= - \int_0^{-1} \left[\frac{-\mu}{G(\mu)} 2 \int_{-1}^1 \Gamma(\mu' \rightarrow \mu) [f_{\text{dir}} S_0 \delta(\mu' - \mu_0) + (1 - f_{\text{dir}}) S_d] \exp(K' L) d\mu' \right] d\mu \\
&= 2 f_{\text{dir}} S_0 \exp(G(\mu_0) L / \mu_0) \int_0^1 \Gamma(\mu_0 \rightarrow -\mu) \frac{\mu}{G(\mu)} d\mu \\
&+ 2(1 - f_{\text{dir}}) S_d \int_0^1 \exp(G(\mu') L / \mu') d\mu' \int_0^1 \Gamma(\mu' \rightarrow -\mu) \frac{\mu}{G(\mu)} d\mu \\
&= f_{\text{dir}} S_0 \exp(G(\mu_0) L / \mu_0) \gamma_0^-(\mu_0) + (1 - f_{\text{dir}}) S_d \gamma_d^-(L)
\end{aligned}$$

where $\gamma_0^-(\mu_0) = 2 \int_0^1 \Gamma(\mu_0 \rightarrow -\mu) \frac{\mu}{G(\mu)} d\mu$, and,

$$\gamma_d^-(L) = 2 \int_0^1 \exp(G(\mu') L / \mu') d\mu' \int_0^1 \Gamma(\mu' \rightarrow -\mu) \frac{\mu}{G(\mu)} d\mu.$$

$$\begin{aligned}
& \int_0^1 \left[\frac{|\mu|}{G(\mu)} 2 \int_{-1}^1 \Gamma(\mu' \rightarrow \mu) I^c(L, \mu') d\mu' \right] d\mu \\
&= 2 \int_0^1 \frac{\mu}{G(\mu)} d\mu \left[\int_0^1 \Gamma(\mu' \rightarrow \mu) I^c(L, \mu') d\mu' + \int_{-1}^0 \Gamma(\mu' \rightarrow \mu) I^c(L, \mu') d\mu' \right] \\
&= 2 \int_0^1 \frac{\mu}{G(\mu)} d\mu \left[\int_0^1 \Gamma(\mu' \rightarrow \mu) I^c(L, \mu') d\mu' + 2 \int_0^1 \Gamma(-\mu' \rightarrow \mu) I^c(L, -\mu') d\mu' \right] \\
&= h_+^+ I^c(L) \uparrow + h_+^- I^c(L) \downarrow,
\end{aligned}$$

where

$$h_+^+ = \frac{2 \int_0^1 \frac{\mu}{G(\mu)} d\mu \int_0^1 \Gamma(\mu' \rightarrow \mu) I^c(L, \mu') d\mu'}{\int_0^1 \mu I^c(L, \mu) d\mu}, \text{ and,}$$

$$h_+^- = \frac{2 \int_0^1 \frac{\mu}{G(\mu)} d\mu \int_0^1 \Gamma(-\mu' \rightarrow \mu) I^c(L, -\mu') d\mu'}{\int_0^1 \mu I^c(L, \mu) d\mu}.$$

$$\begin{aligned}
& \int_{-1}^0 \left[\frac{|\mu|}{G(\mu)} 2 \int_{-1}^1 \Gamma(\mu' \rightarrow \mu) I^c(L, \mu') d\mu' \right] d\mu \\
&= 2 \int_{-1}^0 \frac{-\mu}{G(\mu)} d\mu \left[\int_0^1 \Gamma(\mu' \rightarrow \mu) I^c(L, \mu') d\mu' + 2 \int_{-1}^0 \Gamma(\mu' \rightarrow \mu) I^c(L, \mu') d\mu' \right] \\
&= 2 \int_0^1 \frac{\mu}{G(\mu)} d\mu \left[\int_0^1 \Gamma(\mu' \rightarrow -\mu) I^c(L, \mu') d\mu' + 2 \int_0^1 \Gamma(-\mu' \rightarrow -\mu) I^c(L, -\mu') d\mu' \right] \\
&= h_+^+ I^c(L) \uparrow + h_-^- I^c(L) \downarrow,
\end{aligned}$$

where

$$\begin{aligned}
h_+^+ &= \frac{2 \int_0^1 \frac{\mu}{G(\mu)} d\mu \int_0^1 \Gamma(\mu' \rightarrow -\mu) I^c(L, \mu') d\mu'}{\int_{-1}^0 |\mu| I^c(L, \mu) d\mu}, \text{ and,} \\
h_-^- &= \frac{2 \int_0^1 \frac{\mu}{G(\mu)} d\mu \int_0^1 \Gamma(-\mu' \rightarrow -\mu) I^c(L, -\mu') d\mu'}{\int_{-1}^0 |\mu| I^c(L, \mu) d\mu}.
\end{aligned}$$

So equation (19) can be rewritten as

$$\begin{aligned}
& -\bar{\mu} \frac{\partial}{\partial L} I^c(L) \uparrow + I^c(L) \uparrow = h_+^+ I^c(L) \uparrow + h_+^- I^c(L) \downarrow \\
& + f_{\text{dir}} S_0 \exp(G(\mu_0)L / \mu_0) \gamma_0^+(\mu_0) + (1 - f_{\text{dir}}) S_d \gamma_d^+(L), \\
& \bar{\mu} \frac{\partial}{\partial L} I^c(L) \downarrow + I^c(L) \downarrow = h_-^+ I^c(L) \uparrow + h_-^- I^c(L) \downarrow \\
& + f_{\text{dir}} S_0 \exp(G(\mu_0)L / \mu_0) \gamma_0^-(\mu_0) + (1 - f_{\text{dir}}) S_d \gamma_d^-(L).
\end{aligned}$$

5. The two-stream approximation of radiative transfer model

Let treat the individual leaves as isotropic scattering elements, and make the following identification.

$$\begin{aligned}
h_+^+ &= h_-^- = (1 - \beta)\omega, \\
h_+^- &= h_-^+ = \beta\omega, \\
S_0 \gamma_0^+(\mu_0) &= \bar{\omega} \bar{\mu} \beta_0 G(\mu_0) / \mu_0 = \bar{\omega} \bar{\mu} K_0 \beta_0, K_0 = G(\mu_0) / \mu_0, \\
S_0 \gamma_0^-(\mu_0) &= \bar{\omega} \bar{\mu} (1 - \beta_0) G(\mu_0) / \mu_0 = \bar{\omega} \bar{\mu} K_0 (1 - \beta_0).
\end{aligned}$$

The ω is single leaf albedo, β and β_0 are the upscattering parameters for the diffuse and direct beams, respectively. $\bar{\mu}$ is the average inverse diffuse optical depth per unit leaf area and is approximated by $\bar{\mu} = \int_0^1 \mu / G(\mu) d\mu$. The detailed definition of other parameters can be found in related reference.

Here let ignore the diffuse source S_d , that is, $f_{dir}=1$, the above equation becomes

$$\begin{aligned} -\bar{\mu} \frac{\partial}{\partial L} I^c(L) \uparrow + I^c(L) \uparrow - (1-\beta)\omega I^c(L) \uparrow - \beta\omega I^c(L) \downarrow &= f_{dir} S_0 \exp(G(\mu_0)L / \mu_0) \gamma_0^+(\mu_0) \\ \Rightarrow -\bar{\mu} \frac{\partial}{\partial L} I^c(L) \uparrow + [1 - (1-\beta)\omega] I^c(L) \uparrow - \beta\omega I^c(L) \downarrow &= \bar{\mu} K_0 \beta_0 \exp(K_0 L), \end{aligned} \quad (19)$$

$$\begin{aligned} \bar{\mu} \frac{\partial}{\partial L} I^c(L) \downarrow + I^c(L) \downarrow - h^+ I^c(L) \uparrow - h^- I^c(L) \downarrow &= f_{dir} S_0 \exp(G(\mu_0)L / \mu_0) \gamma_0^-(\mu_0) \\ \Rightarrow \bar{\mu} \frac{\partial}{\partial L} I^c(L) \downarrow + [1 - (1-\beta)\omega] I^c(L) \downarrow &= \bar{\mu} K_0 (1 - \beta_0) \exp(K_0 L). \end{aligned} \quad (20)$$

Equation (19) and (20) are the two-stream approximation radiative transfer model.

References

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